Analysis Report: Differential Expression and Predictive Modeling of COVID-19

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The purpose of this final project was to investigate whether gene expression patterns can effectively distinguish COVID-19 patients from healthy individuals. To accomplish this, publicly available genomic datasets, GSE156063 and GSE152418, were analyzed using statistical and machine learning approaches in RStudio. After cleaning and normalizing the datasets, a two part analysis was conducted. First, differential expression analysis was used to identify genes that changed significantly in response to SARS-CoV-2 infection. Second, Random Forest machine learning models were developed to evaluate whether these genes could accurately classify individuals as either COVID-19 positive or healthy controls.

Differential expression analysis was performed using the DESeq2 and limma packages in R. These tools allowed for the comparison of gene activity levels between COVID-19 patients and healthy individuals. Genes were considered statistically significant if they met two thresholds: an adjusted p-value below 0.05 and a Log2 Fold Change greater than 1.0 or less than -1.0. Positive fold changes represented genes with increased expression in COVID-19 patients, while negative fold changes represented decreased expression.

The analysis identified 10,145 genes that were significantly different between the two groups. This large number of altered genes demonstrates the substantial biological impact that SARS-CoV-2 infection has on the upper airway and immune response. Several highly significant genes, including TK1 and PLK1, were highlighted as examples of genes strongly associated with infection related cellular activity.

To better visualize these results, Volcano Plots were generated using ggplot2 and ggrepel. These plots displayed the relationship between statistical significance and fold change, making it possible to clearly identify genes with the strongest biological differences between healthy and infected samples.

To determine whether these gene expression changes could be used for disease prediction, Random Forest classification models were developed. Random Forest is a machine learning method that uses multiple decision trees to classify data and identify patterns within complex datasets. Several models were tested using different subsets of genes, including the top 10, 25, 50, and 100 most statistically significant genes. The goal was to determine how many genes were necessary to accurately classify COVID-19 patients.

The strongest-performing model used the top 50 genes and achieved 100% accuracy on a test set containing 33 samples. The model also achieved a sensitivity and specificity of 1.0, meaning that every COVID-19 case and every healthy control was classified correctly without false positives or false negatives. Additionally, the p-value associated with the model's accuracy was 3.118×10^{-10}, indicating that the results were extremely unlikely to have occurred by random chance. To evaluate the robustness of the model, another Random Forest model was trained using all available genes rather than a selected subset. Even with thousands of variables included, the model maintained an accuracy of 96.97%. This result demonstrates that the biological signal associated with COVID-19 infection remains strong and detectable across the genome.

In conclusion, this project demonstrates that gene expression analysis is a powerful method for distinguishing COVID-19 patients from healthy individuals. The results show that even a relatively small subset of genes can accurately predict disease status, suggesting that future diagnostic methods may be simplified by focusing on a limited number of key biomarkers rather than requiring full genomic sequencing. Differential expression analysis successfully identified biologically relevant genes, while Random Forest models demonstrated strong predictive capabilities. Although the results were highly accurate, several limitations should be acknowledged. The sample size used in the analysis was relatively small, which may increase the possibility of overfitting. Future work could strengthen these findings through larger datasets, additional validation studies, and cross-validation techniques to ensure broader generalizability.

Tools and Software Used

- R/RStudio - Primary programming and analysis environment
- DESeq2 and limma - Differential expression analysis and normalization
- Random Forest - Predictive machine learning classification
- ggplot2 and ggrepel - Data visualization and Volcano Plot generation